

Week 5 Recitation

Follow-up notes

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Today

Overview

- Recoverability
 - The interpretation of $\overline{FDCH}(Z)$
 - Consistency of the data $\{(p^t, z^t)\}_{t=1}^T$
- Envelope theorem example
 - Roy's identity
- Lagrangian example
- Concave maximization problem example

Envelop Theorem

Consider the following problem:

$$\max_{y \in \mathbb{R}^N} f(y, \theta) \text{ s.t. } g(y, \theta) \leq 0$$

Envelope Theorem:

Assume that

f, g are C^1 functions in $W \times U$,

and for $\forall \theta \in U$, $\exists ! y(\theta) \in W$ that is C^1 , $g(y(\theta), \theta) \geq 0$

Let

- $\mathcal{L}(y, \theta, \lambda)$ be the Lagrangian function,
- $y(\theta), \lambda(\theta)$ solve the KT conditions,
- $V(\theta)$ be the value function

Then $\forall \theta \in U$,

$$\frac{\partial V(\theta)}{\partial \theta} = \left. \frac{\partial \mathcal{L}(y, \theta, \lambda)}{\partial \theta} \right|_{y(\theta), \lambda(\theta)}$$

Envelope Theorem

Why is this useful?:

I want to know what is the effect of a parameter on the welfare of a decision maker

Roy's Identity (Recovering demand)

$$-\frac{\frac{\partial V(p_1, p_2, m)}{\partial p_1}}{\frac{\partial V(p_1, p_2, m)}{\partial m}} = y(p_1, p_2, m)$$

Lagrangian Practice

2022 Quiz 3, #2

$$\begin{aligned} \max_{y \in \mathbb{R}_+^2} \quad & 0.25 \log(y_1) + 0.75 \log(y_2) \\ \text{s.t.} \quad & 5y_1 + 10y_2 \leq 20 \end{aligned}$$

- (a) Set up the Lagrangian function for the optimization problem above. Be sure to include all constraints and put correct sign restrictions on the Lagrange multipliers.
- (b) Are there any constraints that you can drop from the Lagrangian? Justify your answer.

Lagrangian Practice

Fixing one variable

(c) Solve the maximization problem when the Lagrange multiplier on the budget is equal to 2. Solve the Lagrangian finding the optimal value of the Lagrange multiplier. How do the two solutions compare? Explain why they are different or the same.

Determining Concavity

$$f(v|y) = v_1 y_1 + v_2 y_2 + v_3 y_3 - y_1 \ln y_1 - \eta y_2 \ln y_2 - \eta y_3 \ln y_3 - (1 - \eta)(y_2 + y_3) \ln(y_2 + y_3)$$

where $\eta \in \mathbb{R}$

- ① Show the conditional preference function is concave when $\eta \in [0, 1]$.
- ② Suppose $\eta > 1$. Is the preference function concave or no?

- The sum of concave functions is also concave.